GUARD

Graphical Unit for Accident Reporting and Diagnostics

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VISUAL ANALYTICS COURSE

Goal of the visualization

The main purpose is to provide to the members of the Ministry of Infrastructure and Transport a way to identify potential causes of accidents, in order to prevent them. Therefore, our visual analytics solution aims to:

- Provide an overview of national accidents and their characteristics.

- Interactively explore statistical and analytical correlations between multiple factors.

Structure of data

The dataset of accidents occurred in Italy in 2024 was taken from ISTAT.

Attribute	Description
Area	ID of Region
Road section	Number identifying a road section. Identifies: tunnel, bend, crossroad, traffic circle, bump/slope or bottleneck, straight stretch, level crossing.
Accident type	Number identifying the accident type. Identifies: accidents between vehicles, vehicle-pedestrian accidents and accidents involving a single vehicle.
Deadliness	Boolean attribute that identifies if the accident involved death or only injury
Hour of accident	Time of accident measured with hour resolution
Month	Month number
Weekday	Number identifying the weekdays: monday, tuesday, wednesday, etc...
Observations	Number of accidents occurred with the specified attributes

The cleaned final dataset used for the visualization presents 92.402 tuples organized in 8 attributes.

Chosen visualizations

The system is logically divided into four areas:

Geographical distribution:

- Choropleth map
- Bar chart

Temporal distribution:

- Heatmaps

Categorical distribution:

- Bar charts

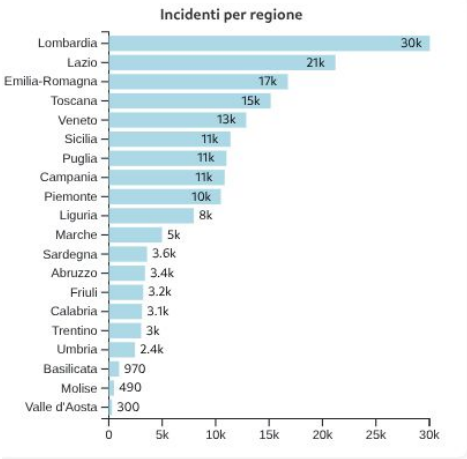
PCA:

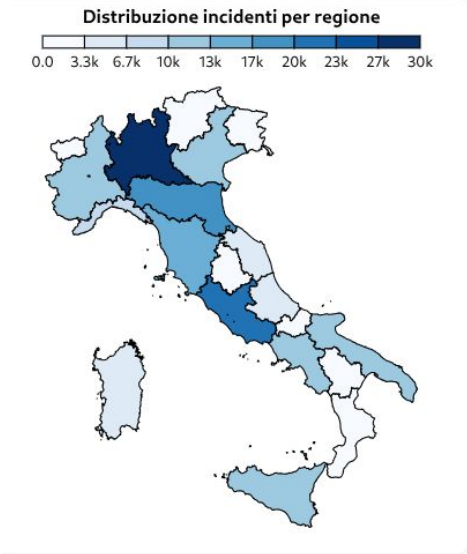
- Scatterplot with color encoding

Geographical distribution

- The map gives a geographical context, making it easier to perceive the spatial correlation between accidents

- The redundant bar chart helps keeping an overview and makes comparisons easier

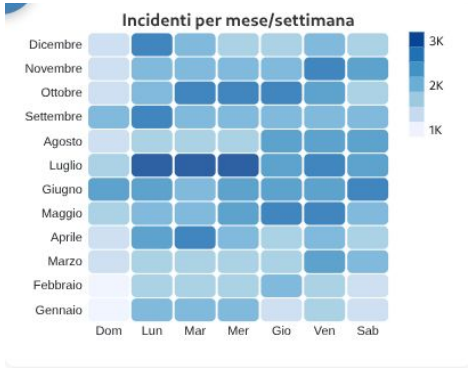


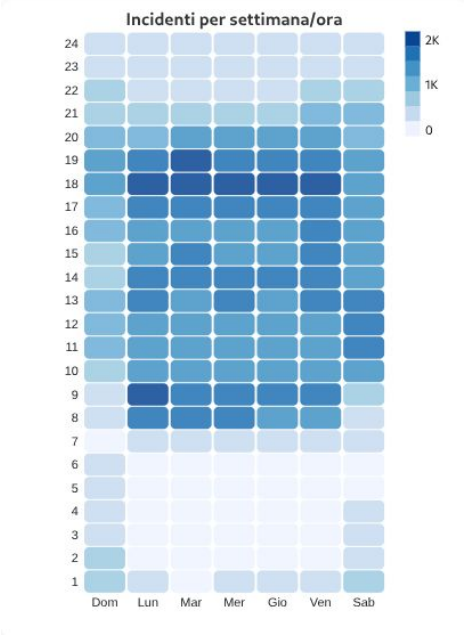


Temporal distribution

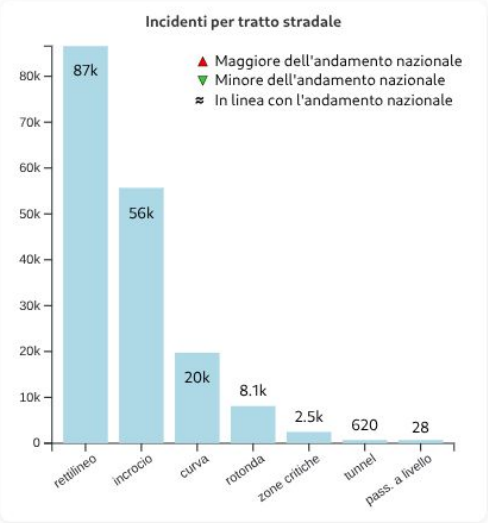
- The first heatmap expresses monthly distribution allowing to analyze seasonal trends.

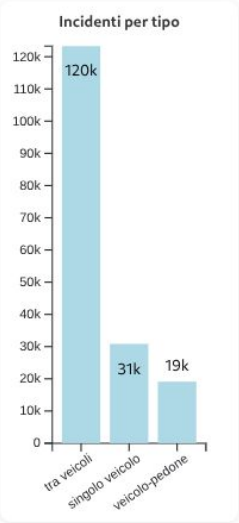
- The latter, analyzes hourly distribution in order to distinguish between daytime and nighttime, as well as highlighting peak commuting hours.

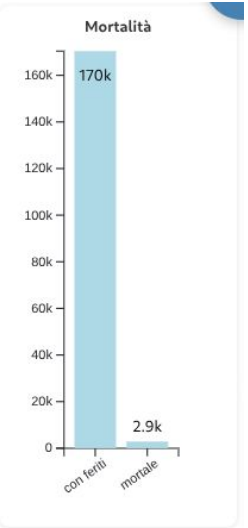




Categorical distribution



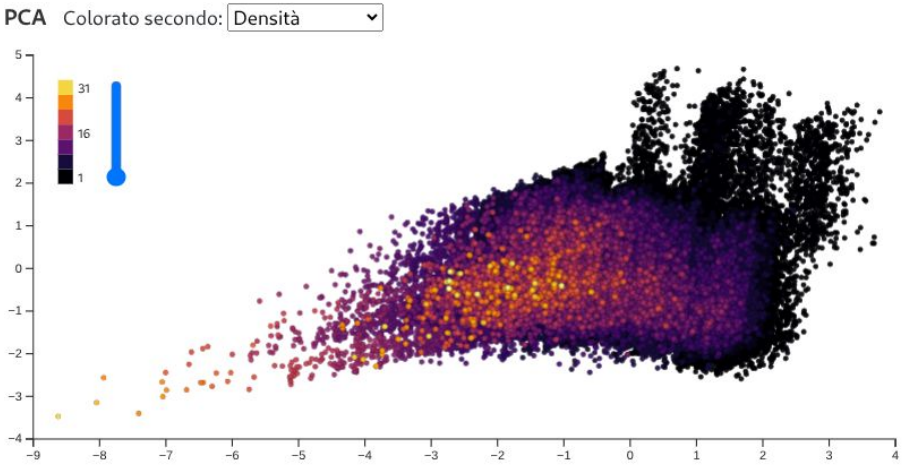




- Sorted bar charts
- Double encoding for more accurate and easier comparisons

PCA

It is encoded to display the distribution according to:



- **Density**: addresses overplotting and highlights similar accidents*
- **Observation**: to recognize accidents sharing common attributes
- **Category**: differentiates accidents by their characteristics

DEMO